

SANDHYA DONGARA

Java Backend Developer

Cincinnati, Ohio | +1 513-268-2872 | sandhya.dongara1@gmail.com

SUMMARY

Java Backend Developer with 6+ years of experience designing and delivering production-grade enterprise applications using Java, J2EE, Spring Boot, and microservices architecture. Began with a strong foundation in core Java, Servlets, JSP, and MVC-based web applications before moving fully into distributed service design with Spring Boot and Spring Cloud. At Citi Bank, spent two years as the backend engineer behind a suite of trade finance analytical services Trade Advisor, Trade Dashboard, LC Life Cycle, Transaction Summary, and Liquidity APIs building REST APIs and event-driven microservices that financial teams relied on daily for trade positions, liquidity visibility, and transaction compliance. Prior to that, contributed to automotive data platforms at BMW Group, leading a legacy-to-microservices migration and optimizing backend data services for performance and maintainability. Consistently, established design patterns MVC, Singleton, Factory, Builder, and Observer to write clean, testable, and maintainable Java code. Comfortable across the full delivery lifecycle: API design, service integration, build automation with Maven and Jenkins, deployment on Linux-based environments via OpenShift, and team collaboration through Jira and Confluence.

TECHNICAL SKILLS

Core Language:	Java (J2EE, Java SE 8+), JSP, Servlets, JDBC
Frameworks:	Spring Boot, Spring MVC, Spring Cloud, Spring Security, Spring AOP, Hibernate, JPA
Architecture:	Microservices, REST API Design, SOAP Web Services, MVC Pattern, Design Patterns (Factory, Singleton, Builder, Observer, Strategy)
API & Integration:	RESTful APIs, SOAP/WSDL, Feign Client, API Gateway, Eureka, Resilience4J, Apache HttpClient
Messaging:	Apache Kafka (producer/consumer, topic design, consumer groups, dead-letter handling)
Security:	Spring Security, OAuth2, JWT, Role-Based Access Control
Databases:	Oracle, PostgreSQL, MySQL, Couchbase; SQL, Stored Procedures, Query Optimization
Build & CI/CD:	Maven, Jenkins, Git, Bitbucket, Docker, Kubernetes, Helm, OpenShift, uDeploy
Web & Server:	JSP, Servlets, Apache Tomcat, HTML, Linux (RHEL/CentOS environment)
Testing & QA:	JUnit, Mockito, REST Assured, SonarQube
Observability:	Kibana, Prometheus, Grafana, Zipkin, Distributed Tracing, Centralized Logging
Collaboration:	Jira, Confluence, Agile/Scrum, Sprint Planning, Code Reviews

PROFESSIONAL EXPERIENCE

Java Backend Developer • Citi Bank (via Tata Consultancy Services)

Feb 2023 – Jan 2025

Domain: Trade Finance Analytics | Financial Data Services | Treasury & Liquidity

Core Service Design & Development

- Designed and built the Trade Advisor and Trade Dashboard backend services using Spring Boot and REST API principles each service following a layered MVC architecture with clearly separated controller, service, and repository tiers. Aggregated trade position data from multiple upstream sources and delivered structured, low-latency JSON responses to analytical front-ends used daily by internal trading teams.

- Engineered the LC (Letter of Credit) Life Cycle microservice to manage state transitions across trade finance instruments. Modeled each lifecycle stage issuance, amendment, and settlement as a domain event published to Kafka topics, making every state change auditable and decoupled from consuming services.
- Built the Transaction Summary Life Cycle service to consolidate high-volume transaction events into unified summary records consumed by reporting and compliance layers. Implemented idempotency checks, retry logic, and dead-letter queue handling to ensure exactly-once processing semantics under sustained Kafka message load.
- Developed Liquidity APIs for treasury teams REST endpoints built on Spring Boot that exposed real-time cash position and fund availability data sourced from Oracle databases. These were latency-sensitive services where response accuracy directly affected treasury decisions; optimized SQL queries and data mapping accordingly.
- Consumed and integrated third-party SOAP-based services for certain trade data feeds, handling XML marshalling, WSDL-based contract definitions, and fault handling within the Spring Web Services layer.

Architecture, Patterns & Integration

- Applied core Java design patterns throughout the codebase Factory for service instantiation, Builder for complex request/response object construction, Singleton for shared infrastructure beans, and Observer for event propagation keeping the services modular and testable across multiple release cycles.
- Structured the microservices ecosystem using Spring Cloud: Eureka for service registry and discovery, Config Server for environment-specific externalized configuration, and API Gateway for centralized routing, rate limiting, and request filtering across all downstream services.
- Wrapped all outbound service calls with Resilience4J circuit breakers and retry policies, preventing cascading failures in a distributed financial environment where downstream service degradation is a routine operational concern.
- Secured all APIs with Spring Security using OAuth2 authorization flows and JWT token validation, applying role-based access control aligned with Citi's enterprise identity and authorization model.
- Contributed to a link migration effort as part of a broader infrastructure modernization program updated service endpoint configurations, re-validated inter-service connectivity, and verified API contract stability across dev, QA, and production environments during the cutover.

Delivery, Deployment & Quality

- Managed application builds using Maven and coordinated CI/CD pipelines through Jenkins automating compilation, test execution, and artifact packaging as part of the standard release workflow.
- Handled deployments on Linux-based infrastructure using OpenShift for containerized releases and uDeploy for environment-specific pipeline management, including manual promotion gates and rollback steps for production deployments.
- Set up structured logging patterns and Kibana dashboards to make service health visible during production runs; log-based triage reduced time-to-diagnosis during incidents without needing direct database access.
- Enforced code quality standards using SonarQube monitored coverage, resolved flagged security hotspots, and cleared technical debt items before merging to the main branch, keeping a multi-team codebase consistently maintainable.
- Wrote JUnit and Mockito unit and integration tests across all owned services, with particular depth on edge cases in transaction state transitions and Kafka consumer behavior under failure conditions.
- Maintained technical documentation and sprint tracking in Confluence and Jira, contributing to team knowledge bases for onboarding and cross-team API integration.

Software Developer • [BMW Group \(via Capgemini\)](#)

Oct 2020 – Feb 2023

Domain: Automotive Data Systems | J2EE Backend | Legacy Modernization

Backend Development & Modernization

- Developed and maintained Java/J2EE backend modules for vehicle configuration, order management, and dealer data services including both Spring Boot microservices and legacy Servlet/JSP-based components deployed on Apache Tomcat. APIs served both internal tooling and customer-facing front ends.
- Led a legacy migration effort that decomposed a tightly coupled J2EE monolith into independently deployable Spring Boot microservices, applying the Strangler Fig pattern to migrate modules incrementally without service disruption. Improved release isolation and reduced cross-team deployment dependency significantly.

- Redesigned critical SQL queries and stored procedures for high-volume automotive reporting datasets eliminated N+1 query patterns, introduced index-aware query structures, and batched large transaction sets, measurably reducing execution time on peak reporting runs.
- Improved REST API response times on data-heavy endpoints by introducing server-side caching, result pagination, and transaction batching, reducing database load during high-concurrency access windows.
- Applied Spring AOP to centralize cross-cutting concerns request/response logging, method-level execution timing, and error capture keeping business logic clean while ensuring consistent instrumentation across all modules.
- Built and maintained Maven-based project structures and Jenkins CI pipelines for automated builds and deployments to staging and production environments via Docker and uDeploy on Linux servers.
- Integrated with external SOAP-based services for vehicle data exchange, managing WSDL contracts and XML payload handling using Spring Web Services and JAXB-based marshalling.
- Tracked development tasks and documented technical decisions in Jira and Confluence; participated in Agile ceremonies including sprint planning, backlog refinement, retrospectives, and peer code reviews.

EDUCATION

M.S. in Information Systems

University of the Cumberlands, Williamsburg, KY

Jan 2025 – May 2026

B.E. in Computer Science & Engineering

S.V. Engineering College for Women, Tirupati, India

Aug 2016 – May 2020

CERTIFICATIONS

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- Oracle Certified Associate, Java SE 8 Programmer